

Telecom Customer Churn Analysis

Data Analytics Portfolio Project

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Tools Used:

- **SQL**
 - **Power BI**
 - **Power Query**
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Project Summary:

This project analyzes telecom customer churn data to identify customer segments with the highest likelihood of churn and the greatest revenue risk. SQL was used for data cleaning and analysis, while Power BI was used to build an interactive dashboard for monitoring churn drivers and customer retention metrics.

Business Problem:

The telecom company is experiencing customer churn and requires a centralized reporting solution to identify retention risks, understand churn drivers, and quantify revenue exposure from customer attrition.

Project Objectives:

Primary Objective

Build a dashboard that helps management understand churn behavior and revenue risk.

Specific Objectives

- Measure overall churn rate
 - Quantify revenue at risk
 - Analyze churn by contract type
 - Analyze churn by internet service type
 - Evaluate customer tenure patterns
 - Identify high-risk customer segment
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Dataset Overview

The dataset contains customer demographic information, account details, service subscriptions, billing information, and churn status.

Key Fields

Field	Description
Customer_ID	Unique customer identifier
Age	Customer age
Contract	Contract type
Internet_Type	Internet service type
Payment_Method	Billing method
Tenure_in_months	Months as customer
Monthly_Charge	Monthly bill amount
Churn	Customer churn status

Key Business Questions

1. What is the overall customer churn rate?
2. Which contract types experience the highest churn rates?
3. Which internet service types have the highest churn rates?
4. How much revenue is at risk due to customer churn?
5. Which customer segments contribute the most revenue at risk?
6. How does customer tenure affect churn behavior?
7. Which customer groups should be targeted for retention initiatives?

<u>KPIs</u>	<u>Value</u>	<u>Description</u>
Total customers:	7k	Total Number of Customers .
Churn Rate:	26.54%	Percentage of Customers who left.
Revenue At Risk:	\$137.09k	Monthly Revenue from churned customers.
Average Tenure:	32.39 Months	Average customer lifespan in months.

SQL Analysis

Key SQL Queries: 6 examples

Total Customers

```
SELECT
    COUNT(DISTINCT Customer_ID) AS TotalCustomers
FROM telecom_customer_churn
```

Churn Rate by Contract:

```
WITH TotalChurn AS
(
    SELECT
        Contract,
        COUNT(DISTINCT Customer_ID) AS Total,
        COUNT(CASE WHEN Customer_Status = 'Churned' THEN 1 END) AS ChurnedCustomers
    FROM telecom_customer_churn
    GROUP BY Contract
)

SELECT
    Contract,
    Total,
    ROUND
    (
        SUM(
            ChurnedCustomers * 100.0 / Total
        )
        ,2
    ) AS ChurnRate
FROM TotalChurn
GROUP BY Contract, Total
```

Churn Rate By Internet Type

```

WITH TotalChurn AS
(
SELECT
    Internet_Type,
    COUNT(DISTINCT Customer_ID) AS Total,
    COUNT(CASE WHEN Customer_Status = 'Churned' THEN 1 END) AS ChurnedCustomers
FROM telecom_customer_churn
GROUP BY Internet_Type
)

SELECT
    Internet_Type,
    ROUND
    (
        SUM(
            ChurnedCustomers * 100.0 / Total
        )
        ,2
    ) AS ChurnRate
FROM TotalChurn
GROUP BY Internet_Type

```

Average Tenure

```

SELECT
    AVG(Tenure_in_months * 1.0) AS AverageTenure
FROM telecom_customer_churn

```

Power BI Dashboard:



Telecom Customer Churn Dashboard

The dashboard allows management to monitor churn performance, identify high-risk customer groups, and measure revenue exposure across customer segments.

KPIS:

1. Total Customers
2. Churn Rate
3. Revenue At Risk
4. Average Tenure

Charts:

1. Churn Rate by Contract
2. Churn Rate by Internet Type
3. Revenue at Risk by Contract
4. Revenue at Risk by Internet Type

Sliders:

1. Contract selector
 2. Internet Type selector
-

Insights:

1. Month-to-Month customers experienced the highest churn rate at approximately 50%, significantly exceeding customers on One-Year and Two-Year contracts.
 2. Fiber Optic customers exhibited the highest churn rate among internet service categories, indicating potential service quality, pricing, or customer expectation issues.
 3. Most of the Revenue at Risk originates from Month-to-Month customers, making this segment the highest retention priority.
 4. Customers with longer tenure generally demonstrate lower churn rates, suggesting that retention improves as customer relationships mature.
 5. A relatively small number of customer segments account for most of the churn-related revenue loss.
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Recommendations:

1. Offer incentives to encourage Month-to-Month customers to transition to longer-term contracts.
2. Investigate factors contributing to elevated churn among Fiber Optic customers.
3. Implement targeted retention campaigns for new customers during their first year.
4. Prioritize retention efforts toward high-value customers contributing the most revenue at risk.
5. Monitor churn trends regularly through dashboard reporting to identify emerging risks.